

2008 ACCESSORIES & BODY, CAB**Instrument Cluster And Panel Illumination - Edge & MKX****DESCRIPTION AND OPERATION****INSTRUMENT CLUSTER AND PANEL ILLUMINATION**

The instrument cluster (IC) and panel illumination is used to provide backlighting to the dimmable and non-dimmable switches and control components.

For Edge vehicles equipped with autolamps, if the exterior lamps are activated during daytime sunload conditions, the message center illumination remains at full intensity and does not dim from the instrument panel dimmer switch during this condition. If the vehicle travels under a bridge or a tunnel, the low level of ambient light causes the illumination level of the message center to change to the level set by the instrument panel dimmer switch. The message center illumination changes back to full intensity when the intense ambient light is restored.

For MKX vehicles equipped with autolamps, if the exterior lamps are activated during daytime sunload conditions, the message center, audio control module (ACM), heating ventilation air conditioning (HVAC) module and switch illumination remain at full intensity and do not dim from the instrument panel dimmer switch during this condition. If the vehicle travels under a bridge or a tunnel, the low level of ambient light causes the illumination level of the message center, ACM, HVAC module and switch to change to the level set by the instrument panel dimmer switch. The message center, ACM, HVAC module and switch illumination change back to full intensity when the intense ambient light is restored.

Dimmable Backlighting

NOTE: **If the battery is disconnected or discharged, the instrument panel dimmer switch requires re-calibration. Rotate the instrument panel dimmer switch from the full dim position to the dome ON position. This will make sure that all displays are visible under all lighting conditions.**

The dimmable switches and components are illuminated when the headlamp switch is in either the PARKING LAMPS ON or HEADLAMPS ON position. The level of intensity is adjustable by rotating the instrument panel dimmer switch either up to increase intensity or down to decrease intensity.

The dimmable switches and components consist of:

- HVAC module - EATC (if equipped)
- HVAC module - EMTC (if equipped)
- Floor shifter
- Heated driver seat switch (if equipped)
- Heated passenger seat switch (if equipped)
- LH rear heated seat switch (if equipped)

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- RH rear heated seat switch (if equipped)
- Message center switch (if equipped)
- Memory set switch (MKX)
- Window adjust switches (MKX)
- Door lock control switches (MKX)
- Headlamp switch
- Instrument panel dimmer switch
- IC
- Speed control switches
- Steering wheel audio control switches
- Hazard/passenger air bag disable/traction control switch
- Audio control module (ACM)
- Power liftgate release switch
- Roof opening panel switch (if equipped)
- Rear entertainment module (RETM) (if equipped)

Non-Dimmable Backlighting

The non-dimmable (fixed) lighting switches and components are illuminated when the delayed accessory relay is energized. The non-dimmable switches and components are illuminated at full lamp intensity.

The non-dimmable lighting switches and components consist of:

- Door lock control switches (Edge)
- Window adjust switches (Edge)

DIAGNOSTIC TESTS

INSTRUMENT CLUSTER AND PANEL ILLUMINATION

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
	Vehicle Communications Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent	-

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 ST2834-A	scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

Principles of Operation

Dimmable Backlighting

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

When the parking lamps are on, the SJB monitors the input from the instrument panel dimmer switch. The SJB sends a pulse-width modulated (PWM) voltage to the dimmable components and switches based on input received from the instrument panel dimmer switch.

The SJB also sends a message over the network to the audio control module (ACM), the rear entertainment module (RETM), the HVAC module - EATC, and the instrument cluster (IC) to indicate the backlighting intensity level. If the receiving module receives invalid backlighting data from the SJB for 5 seconds or less, the receiving module defaults the backlighting to the last setting.

If the receiving module does not receive the backlighting status message from the SJB or if the data received is deemed invalid for more than 5 seconds, the receiving module sets DTC U0140 in continuous memory (except the HVAC module - EATC) and defaults the backlighting to full intensity.

Fixed Backlighting

For Edge, when the accessory delay relay is energized, switched voltage is supplied to the window control and door lock switches.

The MKX is not equipped with any fixed backlighting components.

Field-Effect Transistor (FET) Protection

The SJB utilizes a FET protective circuit strategy for many of its outputs (for example, the LED backlighting output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down (turns off the voltage or ground provided by the module) when a fault is detected. A continuous DTC is stored at that time for the fault. The circuit will then reset after a customer demand of the function (switching the component on, battery saver being energized). When an excessive circuit load occurs several times, the

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module shuts down the output until a repair procedure is carried out. At the same time, the continuous DTC that was stored on the first failure will not clear by a command to clear the continuous DTCs. The module will not allow this code to be cleared or the circuit restored to normal until a successful on-demand self-test proves that the fault has been repaired. After the on-demand self-test has successfully completed (no on-demand DTCs present), the continuous DTC will have been cleared and the circuit function will return.

Each circuit has 3 predefined levels of short circuit tolerance established in the module based on each circuits' capability. When the first or second level is reached, the continuous DTC associated with the circuit sets along with DTC B106E. These DTCs may be cleared using the Clear DTC operation on the scan tool as long as the fault itself has been corrected. If any of the circuits are shorted past the third level, then DTCs B106F and B1342 set along with the associated continuous DTC. These DTCs cannot be cleared and the module must be replaced.

The SJB FET protected output circuits for the instrument panel illumination system are for the white lighting (MKX) and LED backlighting (Edge) output circuits.

Instrument Panel Dimmer Switch Calibration

If the battery is disconnected or discharged, the instrument panel dimmer switch requires re-calibration. Rotate the instrument panel dimmer switch from the full dim position to the dome ON position. This will make sure that all displays are visible under all lighting conditions.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.

VISUAL INSPECTION CHART

Electrical
• Smart junction box (SJB) fuse(s): ○ 10 (15A) (dimnable illumination) ○ 41 (15A) (non-dimnable illumination) • Wiring, terminals or connectors • Illuminated components • SJB

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure to use the latest scan tool release software.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power

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and ground from the DLC are provided to the VCM.

5. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
 - If the scan tool responds with no communication with the SJB, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **SYMPTOM CHART** table.

DTC Charts

SMART JUNCTION BOX (SJB) DTC CHART

DTC	Description	Action
B2212	Panel Dim Switch Out of Range	Go to Pinpoint Test C .
B2A32	LED Backlighting Output Circuit Open	If the backlighting components are inoperative, go to Pinpoint Test A . If the backlighting components are always on, go to Pinpoint Test B .
B2A33	LED Backlighting Output Circuit Short to Ground	Go to Pinpoint Test A .
B29B7	White Lighting Output Circuit Open	If the backlighting components are inoperative, go to Pinpoint Test A . If the backlighting components are always on, go to Pinpoint Test B .
B29B8	White Lighting Output Circuit Short to Ground	Go to Pinpoint Test A .
All other DTCs	-	REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article.

Symptom Chart

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SYMPTOM CHART

Condition	Possible Sources	Action
No communication with the smart junction box (SJB)	- Fuse(s) - Wiring, terminals or connectors - SJB	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
The control illumination is inoperative - all hardwired illumination	- Fuse - Wiring, terminals or connectors - SJB	Go to Pinpoint Test a .
The control illumination is always on - all hardwired illumination	- Wiring, terminals or connectors - SJB	Go to Pinpoint Test B .
All instrument panel illumination does not dim	- Wiring, terminals or connectors - Instrument panel dimmer switch - SJB	Go to Pinpoint Test C .
The steering wheel control illumination is inoperative	- Wiring, terminals or connectors - Steering wheel control switch(es) - Steering wheel control switch harness - Clockspring	Go to Pinpoint Test D .
The window control switch illumination is inoperative	- Wiring, terminals or connectors - Window control switch	- For MKX, go to Pinpoint Test E . - For Edge, CHECK the operation of the power windows. - If the power windows operate correctly, INSTALL a new window control switch. REFER to <u>GLASS, FRAMES AND MECHANISMS</u> article. TEST the system for normal operation. - If the power windows do not operate correctly, REFER to <u>GLASS, FRAMES AND MECHANISMS</u> article. - CARRY OUT the network test.

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- All network-controlled illumination does not dim

- The instrument cluster (IC) illumination is inoperative

- The audio control module (ACM) illumination is inoperative

- Medium speed controller area network (MS-CAN)
- SJB

- IC

- ACM

- If the scan tool responds with no communication to any module on the MS-CAN, REFER to **MODULE COMMUNICATIONS NETWORK** article.
- If the scan tool communicates with the modules on the MS-CAN, INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.
- CHECK the audio control module (ACM) illumination for correct operation.
 - If the ACM illumination operates correctly, INSTALL a new IC. REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES** article. TEST the system for normal operation.
 - If the ACM illumination does not operate correctly, REFER to **MODULE COMMUNICATIONS NETWORK** article to continue diagnosis of the communications network.
- CHECK the IC illumination for correct operation.
 - If the IC illumination operates correctly, REMOVE the ACM. REFER to **INFORMATION AND ENTERTAINMENT SYSTEMS** article. SEND it to an authorized repair facility. TEST the system for normal operation after the repair.
 - If the IC illumination does not operate correctly, REFER to **MODULE COMMUNICATIONS NETWORK** article to continue

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		diagnosis of the communications network.
The rear entertainment module (RETM) illumination is inoperative	RETM	REMOVE the RETM. REFER to <u>INFORMATION AND ENTERTAINMENT SYSTEMS</u> article. SEND it to an authorized repair facility. TEST the system for normal operation.
The HVAC module - EATC illumination is inoperative	HVAC module	- CHECK the IC illumination for correct operation. - If the IC illumination operates correctly, INSTALL a new HVAC module. REFER to <u>CLIMATE CONTROL</u> article. TEST the system for normal operation. - If the IC illumination does not operate correctly, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to continue diagnosis of the communications network.
The individual component illumination is inoperative	- Wiring, terminals or connectors - Floor shifter - LH rear heated seat switch (if equipped) - RH rear heated seat switch (if equipped) - Manual temperature control (if equipped) - Message center switch (if equipped) - Memory set switch (MKX) - Window adjust switches (MKX) - Headlamp switch - Hazard/passenger air bag disable/traction control switch (if equipped)	Go to Pinpoint Test E .

	• Door lock control switches • Power liftgate release switch • Roof opening panel switch (if equipped)	
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Pinpoint Tests

Pinpoint Test A: The Control Illumination Is Inoperative - All Hardwired Illumination

Refer to appropriate SYSTEM WIRING DIAGRAM article for Cluster and Panel Illumination schematic and connector information.

Normal Operation

Based on input received from the instrument panel dimmer switch, the smart junction box (SJB) sends a pulse-width modulated (PWM) voltage to the hardwired dimmable illumination sources through circuit VLN37 (BU/WH) (MKX) or circuit VLN04 (VT/GY) (Edge).

DTC Description	Fault Trigger Conditions
B2A32 - LED Backlighting Output Circuit Open	An on-demand DTC that sets when the SJB detects an open on the LED backlighting output circuit.
B2A33 - LED Backlighting Output Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground on the LED backlighting output circuit. This DTC also sets when fuse 10 (15A) has failed or is removed.
B29B7 - White Lighting Output Circuit Open	An on-demand DTC that sets when the SJB detects an open on the white lighting output circuit.
B29B8 - White Lighting Output Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground on the white lighting output circuit. This DTC also sets when fuse 10 (15A) has failed or is removed.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- SJB

PINPOINT TEST A: THE CONTROL ILLUMINATION IS INOPERATIVE - ALL HARDWIRED ILLUMINATION

A1 CHECK FOR RECORDED SJB DTCs FROM THE ON-DEMAND SELF-TEST

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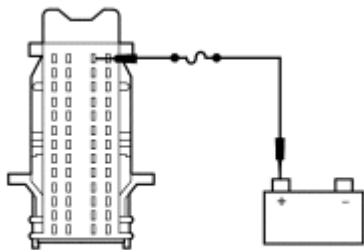
- Use the recorded results from the SJB self-test.
- **Are any DTCs recorded?**
YES : For DTC B2A33, B2A32, B29B7 or B29B8, go to A3.
NO : Go to A2.

A2 CHECK THE INSTRUMENT PANEL DIMMER SWITCH

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: SJB DataLogger
- Monitor the SJB PID (DIM_SW_3) while rotating the instrument panel dimmer switch from the lowest setting to the highest setting.
- **Does the PID indicate the switch is operating correctly?**
YES : Go to A4.
NO : INSTALL a new instrument panel dimmer switch. REFER to **INSTRUMENT PANEL DIMMER SWITCH**. TEST the system for normal operation.

A3 CHECK CIRCUIT VLN37 (BU/WH) OR CIRCUIT VLN04 (VT/GY) FOR AN OPEN

- Key in OFF position.
- Disconnect: SJB C2280b

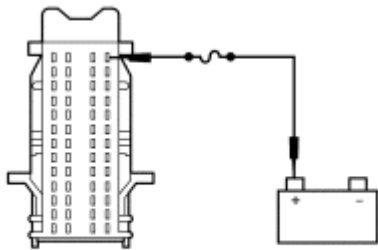


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Fig. 1: Connecting Fused (5A) Jumper Wire Between SJB C2280B-39, Circuit VLN04 (VT/GY) & Battery Positive
Courtesy of FORD MOTOR CO.

NOTE: If the jumper wire fails, repair circuit VLN04 (VT/GY) for a short to ground.

- For Edge, connect a fused (5A) jumper wire between the SJB C2280b-39, circuit VLN04 (VT/GY) and battery positive.



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Fig. 2: Connecting Fused (5A) Jumper Wire Between SJB C2280B-52, Circuit VLN37 (BU/WH) & Battery Positive
Courtesy of FORD MOTOR CO.

NOTE: If the jumper wire fails, repair circuit VLN37 (BU/WH) for a short to ground.

- For MKX, connect a fused (5A) jumper wire between the SJB C2280b-52, circuit VLN37 (BU/WH) and battery positive.

- **Do the backlighting components illuminate?**

YES : REMOVE the jumper wire. Go to A4.

NO : REMOVE the jumper wire. VERIFY the BJB fuse 10 (15A) is OK. REPAIR the circuit in question for a short to ground. RUN the on-demand self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

A4 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. RUN the on-demand self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

Pinpoint Test B: The Control Illumination Is Always On - All Hardwired Illumination

Refer to appropriate SYSTEM WIRING DIAGRAM article for Cluster and Panel Illumination schematic and connector information.

Normal Operation

The smart junction box (SJB) supplies a pulse-width modulated (PWM) voltage signal to the dimmable illumination sources on circuit VLN04 (VT/GY) (Edge) and VLN37 (BU/WH) (MKX).

- DTC B2A32 (LED Backlighting Output Circuit Open) - is an on-demand DTC that sets when the SJB detects a short to voltage on the LED backlighting output circuit.
- DTC B29B7 (White Lighting Output Circuit Open) - is an on-demand DTC that sets when the SJB detects a short to voltage on the white lighting output circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- SJB

PINPOINT TEST B: THE CONTROL ILLUMINATION IS ALWAYS ON - ALL HARDWIRED ILLUMINATION

B1 CHECK THE EXTERIOR LIGHTING

- Key in OFF position.
- Place the headlamp switch in the OFF position.
- Observe the exterior lighting components.
- Key in ON position.
- **Are any exterior lighting components illuminated?**
YES : REFER to **EXTERIOR LIGHTING** article.
NO : Go to B2.

B2 CHECK CIRCUITS VLN04 (VT/GY) (EDGE) AND VLN37 (BU/WH) (MKX) FOR A SHORT TO VOLTAGE

- Disconnect: SJB C2280b
- Key in ON position.
- **Do the dimmable backlit components continue to illuminate?**
YES : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.
NO : Go to B3.

B3 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

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- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: All Instrument Panel Illumination Does Not Dim

Refer to appropriate SYSTEM WIRING DIAGRAM article for Cluster and Panel Illumination schematic and connector information.

Normal Operation

The smart junction box (SJB) monitors the input from the instrument panel dimmer switch by sending a voltage reference signal through circuit VLN18 (BU/WH) to the instrument panel dimmer switch. The return signal is sent through circuit CLN27 (WH/BN) back to the SJB. Based on the input from the instrument panel dimmer switch, the SJB sends a pulse-width modulated (PWM) voltage signal to the dimmable illumination sources. If the SJB detects a concern with the circuitry to the instrument panel dimmer switch, the SJB defaults the backlighting to a full intensity level.

The SJB also sends a message over the network to the audio control module (ACM), the rear entertainment module (RETM), the HVAC module - EATC, and the instrument cluster (IC) to indicate the backlighting intensity level. If the receiving module receives invalid backlighting data from the SJB for 5 seconds or less, the receiving module defaults the backlighting to the last setting.

If the receiving module does not receive the backlighting status message from the SJB or if the data received is deemed invalid for more than 5 seconds, the receiving module sets DTC U0140 in continuous memory (except the climate control module) and defaults the backlighting to full intensity.

- DTC B2212 (Panel Dim Switch Out Of Range) - is an on-demand DTC that sets when the SJB detects an out of range condition on the dimmer input switch circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals, or connectors
- Instrument panel dimmer switch
- SJB

PINPOINT TEST C: ALL INSTRUMENT PANEL ILLUMINATION DOES NOT DIM

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Calibrate the instrument panel dimmer switch by rotating the dimmer switch from the full dim position to the dome ON position before continuing

diagnostics.**C1 CHECK THE EXTERIOR LIGHTING OPERATION**

- Place the headlamp switch in each of its positions while observing the exterior lamps.
- **Do the exterior lamps operate correctly?**

YES : Go to C2.

NO : REFER to **EXTERIOR LIGHTING** article.

C2 CHECK THE RECORDED RESULTS FROM BOTH CONTINUOUS AND ON-DEMAND SELF-TESTS

- Check for recorded DTCs from the continuous and on-demand self-tests.
 - **Are any DTCs present?**
- YES** : If DTC B2212 is retrieved, go to C4.

For all other DTCs, REFER to **MULTIFUNCTION ELECTRONIC MODULES** article.

NO : Go to C3.

C3 CHECK THE INSTRUMENT PANEL DIMMER SWITCH INPUT

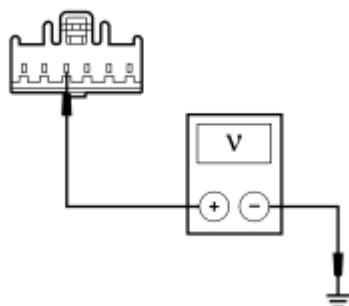
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: SJB DataLogger
- Monitor the SJB PID (DIM_SW_3) while rotating the instrument panel dimmer switch from the lowest setting to the highest setting.
- **Does the PID indicate the switch is operating correctly?**

YES : Go to C9.

NO : INSTALL a new instrument panel dimmer switch. REFER to **INSTRUMENT PANEL DIMMER SWITCH**.

C4 CHECK CIRCUIT CLN27 (WH/BN) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: SJB C2280b
- Disconnect: Instrument Panel Dimmer Switch C2298
- Key in ON position.



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Fig. 3: Measuring Voltage Between Instrument Panel Dimmer Switch C2298-4, Circuit

CLN27 (WH/BN) & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between the instrument panel dimmer switch C2298-4, circuit CLN27 (WH/BN), harness side and ground.
- **Is any voltage indicated?**
YES : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
NO : Go to C5.

C5 CHECK CIRCUIT VLN18 (BU/WH) FOR A SHORT TO VOLTAGE

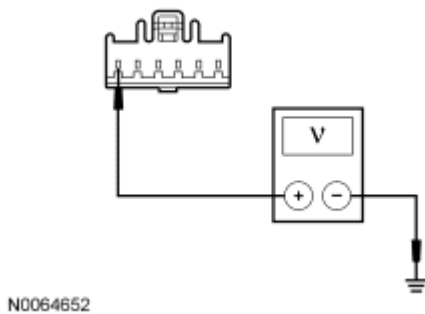


Fig. 4: Measuring Voltage Between Instrument Panel Dimmer Switch C2298-6, Circuit VLN18 (BU/WH) & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between the instrument panel dimmer switch C2298-6, circuit VLN18 (BU/WH), harness side and ground.
- **Is any voltage indicated?**
YES : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
NO : Go to C6.

C6 CHECK CIRCUIT CLN27 (WH/BN) FOR AN OPEN OR A SHORT TO GROUND

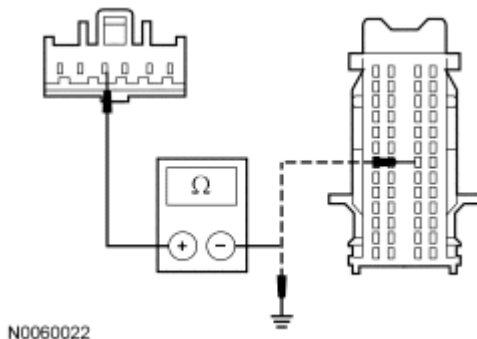


Fig. 5: Measuring Resistance Between Instrument Panel Dimmer Switch C2298-4, Circuit CLN27 (WH/BN) & SJB C2280B-33
Courtesy of FORD MOTOR CO.

- Measure the resistance between the instrument panel dimmer switch C2298-4, circuit CLN27 (WH/BN), harness side and the SJB C2280b-33, circuit CLN27 (WH/BN), harness side; and between the instrument panel dimmer switch C2298-4, circuit CLN27 (WH/BN), harness side and ground.
- **Is the resistance less than 5 ohms between the instrument panel dimmer switch and the SJB, and greater than 10,000 ohms between the instrument panel dimmer switch and ground?**
YES : Go to C7.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

C7 CHECK CIRCUIT VLN18 (BU/WH) FOR AN OPEN OR A SHORT TO GROUND

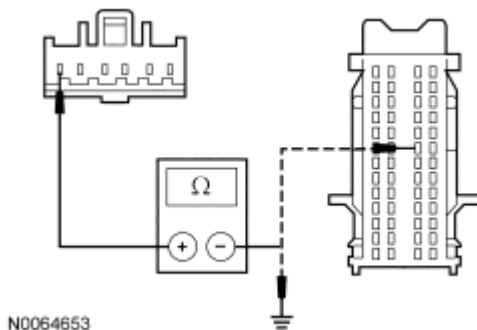


Fig. 6: Measuring Resistance Between Instrument Panel Dimmer Switch C2298-6, Circuit VLN18 (BU/WH) & SJB C2280B-34
 Courtesy of FORD MOTOR CO.

- Measure the resistance between the instrument panel dimmer switch C2298-6, circuit VLN18 (BU/WH), harness side and the SJB C2280b-34, circuit VLN18 (BU/WH), harness side; and between the instrument panel dimmer switch C2298-6, circuit VLN18 (BU/WH), harness side and ground.
- **Is the resistance less than 5 ohms between the instrument panel dimmer switch and the SJB, and greater than 10,000 ohms between the instrument panel dimmer switch and ground?**
YES : Go to C8.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

C8 CHECK THE INSTRUMENT PANEL DIMMER SWITCH

- Carry out the component test for the instrument panel dimmer switch. Refer to COMPONENT TESTING.
- **Is the instrument panel dimmer switch OK?**
YES : Go to C9.
NO : INSTALL a new instrument panel dimmer switch. REFER to **INSTRUMENT PANEL DIMMER SWITCH**. TEST the system for normal operation.

C9 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion

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- damaged pins
- pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test D: The Steering Wheel Control Illumination Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAM article for Cluster and Panel Illumination schematic and connector information.

Normal Operation

The smart junction box (SJB) sends a pulse width modulated (PWM) voltage to the steering wheel controls through circuit VLN04 (VT/GY) (Edge) or circuit VLN37 (BU/WH) (MKX) to the clockspring. The voltage passes through the clockspring to the steering wheel control switches. Ground to the clockspring is supplied through circuit GD116 (BK/VT).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Steering wheel control switch(es)
- Steering wheel control switch harness
- Clockspring

PINPOINT TEST D: THE STEERING WHEEL CONTROL ILLUMINATION IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

D1 CHECK THE OTHER DIMMABLE LIGHTING SWITCHES

- Key in OFF position.
- Place the headlamp switch in the PARKING LAMPS ON position.
- Rotate the instrument panel dimmer switch to the full illumination position and observe other dimmable switches (door lock switches, hazard lamp switch, headlamp switch for example).
- **Are all the dimmable switches inoperative?**

YES : Go to **Pinpoint Test A** .

NO : Go to D2.

D2 CHECK CIRCUIT VLN04 (VT/GY) (EDGE) OR CIRCUIT VLN37 (BU/WH) (MKX) FOR AN OPEN

- Place the headlamp switch in the OFF position.
- Depower the supplemental restraint system (SRS). Refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
- Disconnect: Clockspring C218a
- Connect the battery ground cable.
- Place the headlamp switch in the PARKING LAMPS ON position.
- Rotate the instrument panel dimmer switch to the full illumination position.

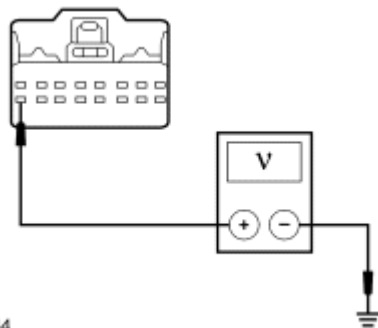


Fig. 7: Measuring Voltage Between Clockspring C218A-16, Circuit VLN04 (VT/GY) (Edge) Or Circuit VLN37 (BU/WH) (MKX)
Courtesy of FORD MOTOR CO.

- Measure the voltage between the clockspring C218a-16, circuit VLN04 (VT/GY) (Edge) or circuit VLN37 (BU/WH) (MKX), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : TURN the parking lamps off. Go to D3.
NO : REPAIR the circuit. REPOWER the SRS. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

D3 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN

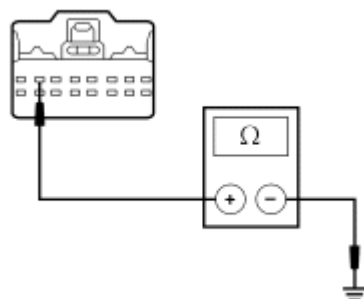


Fig. 8: Measuring Resistance Between Clockspring C218A-7, Circuit GD116 (BK/VT) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218a-7, circuit GD116 (BK/VT), harness side and ground.

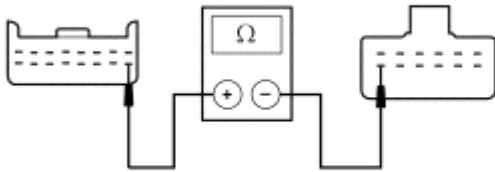
- **Is the resistance less than 5 ohms?**

YES : Go to D4.

NO : REPAIR the circuit. REPOWER the SRS. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

D4 CHECK THE CLOCKSPRING FOR AN OPEN

- Remove the driver air bag module. Refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
- Disconnect: Upper Clockspring Connector C218b

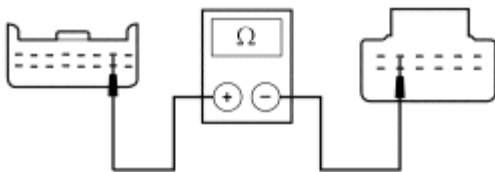


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Fig. 9: Measuring Resistance Between Clockspring C218A Pin 16 And Upper Clockspring C218B Pin 7

Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218a pin 16, component side and the upper clockspring C218b pin 7, component side.



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Fig. 10: Measuring Resistance Between Clockspring C218A Pin 7 And Upper Clockspring C218B Pin 2

Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218a pin 7, component side and the upper clockspring C218b pin 2, component side.

- **Are the resistances less than 5 ohms?**

YES : Go to D5.

NO : INSTALL a new clockspring. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

D5 CHECK THE STEERING WHEEL HARNESS CIRCUIT VLN04 (VT/GY) (EDGE) OR CIRCUIT VLN37 (BU/WH) (MKX) FOR AN OPEN

- Disconnect: LH Steering Wheel Switches C2998
- Disconnect: RH Steering Wheel Switches C2999

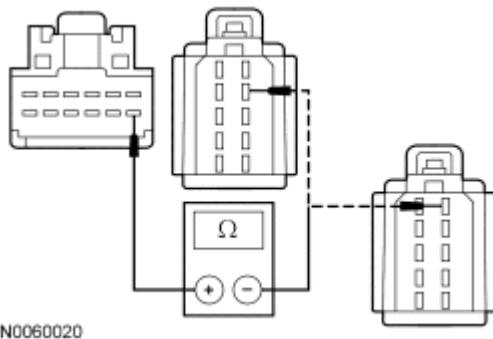


Fig. 11: Measuring Resistance Between Upper Clockspring Connector C218B-7, Circuit VLN04 (VT/GY) & LH Steering Wheel Switches C2998-2
Courtesy of FORD MOTOR CO.

- Measure the resistance between the upper clockspring connector C218b-7, circuit VLN04 (VT/GY), harness side and the LH steering wheel switches C2998-2, circuit VLN04 (VT/GY), harness side; and between the upper clockspring connector C218b-7, circuit VLN04 (VT/GY), harness side and the RH steering wheel switches C2999-1, circuit VLN04 (VT/GY), harness side.
- **Are the resistances less than 5 ohms?**

YES : Go to D6.

NO : REPAIR the circuit in question. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

D6 CHECK CIRCUIT GD165 (BK) FOR AN OPEN

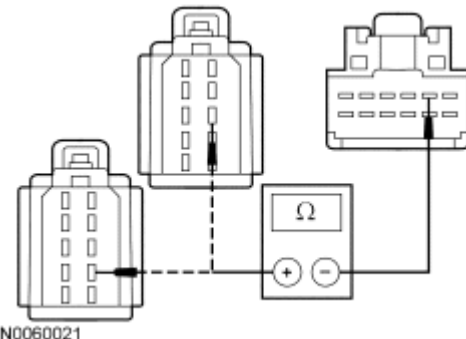


Fig. 12: Measuring Resistance Between Clockspring C218B-2, Circuit GD165 (BK) & LH Steering Wheel Switch C2998-4
Courtesy of FORD MOTOR CO.

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- Measure the resistance between the clockspring C218b-2, circuit GD165 (BK), harness side, and the LH steering wheel switch C2998-4, circuit GD165 (BK), harness side; and between the clockspring C218b-2, circuit GD165 (BK), harness side, and the RH steering wheel switch C2999-3, circuit GD165 (BK), harness side.

- **Is the resistance less than 5 ohms?**

YES : INSTALL a new speed control switch or steering wheel controls as necessary. REFER to **SPEED CONTROL** article or **INFORMATION AND ENTERTAINMENT SYSTEMS** article. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

NO : REPAIR the circuit. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

Pinpoint Test E: The Individual Component Illumination Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAM article for Cluster and Panel Illumination schematic and connector information.

Normal Operation

The smart junction box (SJB) sends a pulse-width modulated (PWM) voltage to the dimmable components through circuit VLN04 (VT/GY) (Edge) or circuit VLN37 (BU/WH) (MKX). Ground is provided through circuits GD116 (BK/VT), GD126 (BK/WH) and GD139 (BK/YE).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Floor shifter
- LH rear heated seat switch (if equipped)
- RH rear heated seat switch (if equipped)
- Manual temperature control (if equipped)
- Message center switch (if equipped)
- Memory set switch (MKX)
- Window adjust switches (MKX)
- Door lock control switches
- Headlamp switch
- Hazard/passenger air bag disable/traction control switch (if equipped)
- Power liftgate release switch
- Roof opening panel switch (if equipped)

PINPOINT TEST E: THE INDIVIDUAL COMPONENT ILLUMINATION IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to

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use the correct probe adapter(s) may damage the connector.

E1 CHECK THE VOLTAGE TO THE SINGLE ILLUMINATION SOURCE

- Key in OFF position.
- Disconnect: Inoperative Illumination Source
- Key in ON position.
- With the parking lamps on, rotate the instrument panel dimmer switch to the full illumination position.
- Measure the voltage between the inoperative illumination source, harness side and ground as follows:

EDGE

Component	Connector-Pin	Circuit
Floor shifter	C3245-3	VLN04 (VT/GY)
Manual climate control	C2357a-4	VLN04 (VT/GY)
Headlamp switch	C205-5	VLN04 (VT/GY)
Message center switch	C253-1	VLN04 (VT/GY)
Hazard/passenger air bag disable/traction control switch	C2355-6	VLN04 (VT/GY)
LH door lock control switch	C505-2	CBP41 (BU)
RH door lock control switch	C605-2	CBP41 (BU)
Power liftgate release switch	C2269-3	VLN04 (VT/GY)

MKX

Component	Connector-Pin	Circuit
Floor shifter	C3245-3	VLN37 (BU/WH)
Memory set switch	C503-3	VLN37 (BU/WH)
Headlamp switch	C205-5	VLN37 (BU/WH)
Message center	C253-1	VLN37

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switch		(BU/WH)
Hazard/passenger air bag disable/traction control switch	C2355-6	VLN37 (BU/WH)
Power liftgate release switch	C2269-3	VLN37 (BU/WH)
Roof opening panel switch	C912-2	VLN37 (BU/WH)
LH front window control switch	C535a-6	VLN37 (BU/WH)
RH front window control switch	C648-1	VLN37 (BU/WH)
LH rear window control switch	C701-4	VLN37 (BU/WH)
RH rear window control switch	C801-4	VLN37 (BU/WH)
LH door lock control switch	C505-2	VLN37 (BU/WH)
RH door lock control switch	C605-2	VLN37 (BU/WH)
LH rear heated seat switch	C722-5	VLN37 (BU/WH)
RH rear heated seat switch	C822-5	VLN37 (BU/WH)

- **Is the voltage greater than 10 volts?**

YES : For LH or RH rear window switches, **INSTALL** a new switch for the switch in question. REFER to **GLASS, FRAMES AND MECHANISMS** article.

For all other components, go to E2.

NO : REPAIR the circuit in question. TEST the system for normal operation.

E2 CHECK ILLUMINATION SOURCE GROUND CIRCUIT FOR AN OPEN

- Key in OFF position.
- Measure the resistance between the inoperative illumination source, harness side and ground as follows:

EDGE

Component	Connector-Pin	Circuit
Floor shifter	C3245-4	GD140 (BK/GN)
Manual climate		GD115

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control	C2357a-23	(BK/GY)
Headlamp switch	C205-10	GD116 (BK/VT)
Message center switch	C253-4	GD116 (BK/VT)
Hazard/passenger air bag disable/traction control switch	C2355-1	GD116 (BK/VT)
LH door lock control switch	C505-4	GD133 (BK)
RH door lock control switch	C605-4	GD140 (BK/GN)
Power liftgate release switch	C2269-2	GD116 (BK/VT)

MKX

Component	Connector-Pin	Circuit
Floor shifter	C3245-4	GD140 (BK/GN)
Memory set switch	C503-7	GD133 (BK)
Headlamp switch	C205-10	GD116 (BK/VT)
Message center switch	C253-4	GD116 (BK/VT)
Hazard/pad/traction control switch	C2355-1	GD116 (BK/VT)
Power liftgate release switch	C2269-2	GD116 (BK/VT)
Roof opening panel switch	C912-1	GD133 (BK)
LH front window control switch	C535b-6	GD133 (BK)
RH front window control switch	C648-6	GD140 (BK/GN)
LH door lock control switch	C505- 4	GD133 (BK)
RH door lock control switch	C605-4	GD140 (BK/GN)
LH rear heated seat switch	C722-1	GD126 (BK/WH)
RH rear heated seat		GD145

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switch

C822-1

(BK/BU)

- Is the resistance less than 5 ohms?
YES : INSTALL a new component in question. TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

REMOVAL AND INSTALLATION

INSTRUMENT PANEL DIMMER SWITCH

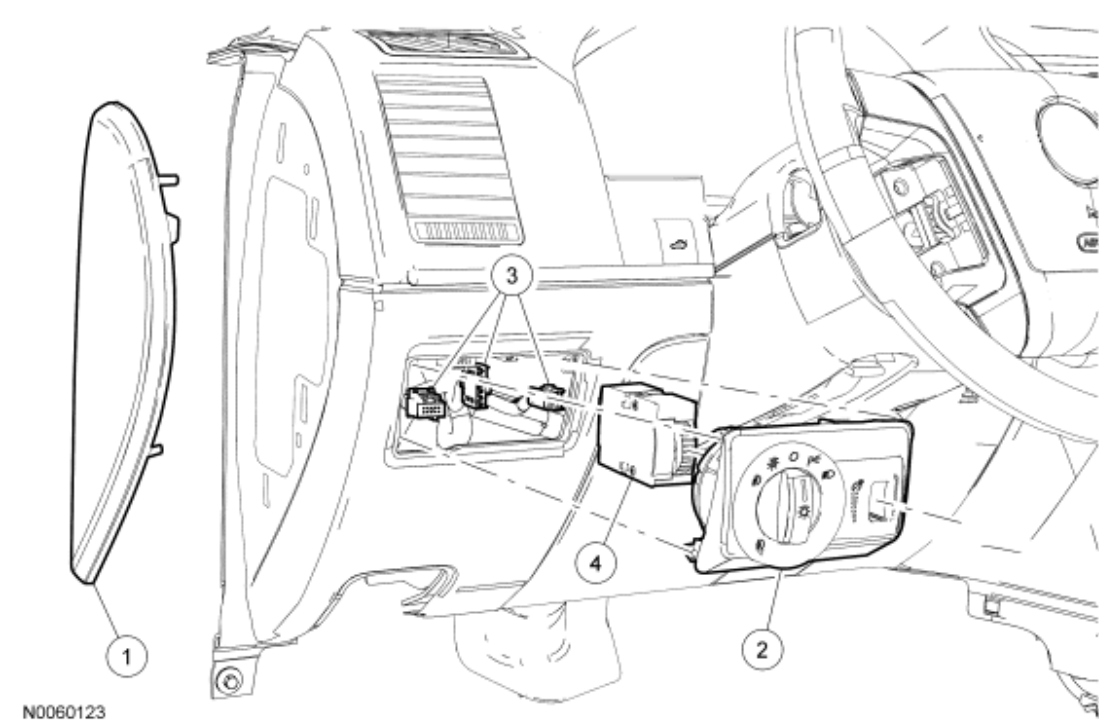


Fig. 13: Exploded View Of Instrument Panel Dimmer Switch
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	5404481	Instrument panel side finish panel
2	14K147	Headlamp switch assembly
3	-	Headlamp switch assembly electrical connectors (part of 14401)
4	11691	Instrument panel dimmer switch

REMOVAL AND INSTALLATION

1. Remove the instrument panel side finish panel.

NOTE: The switch is removed by pushing from behind.

2. Release the tabs and remove the headlamp switch assembly.
 - Disconnect the electrical connectors.
3. Release the 4 tabs and remove the instrument panel dimmer switch.

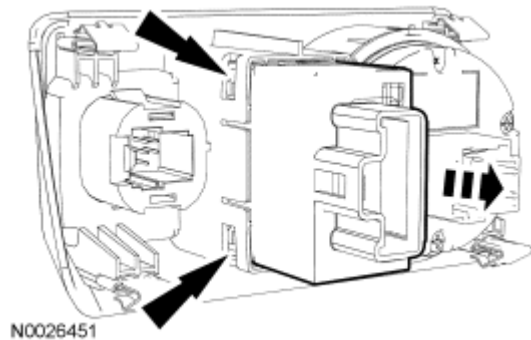


Fig. 14: Releasing Tabs & Removing Instrument Panel Dimmer Switch
Courtesy of FORD MOTOR CO.

4. To install, reverse the removal procedure.

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SPECIFICATIONS

TORQUE SPECIFICATIONS

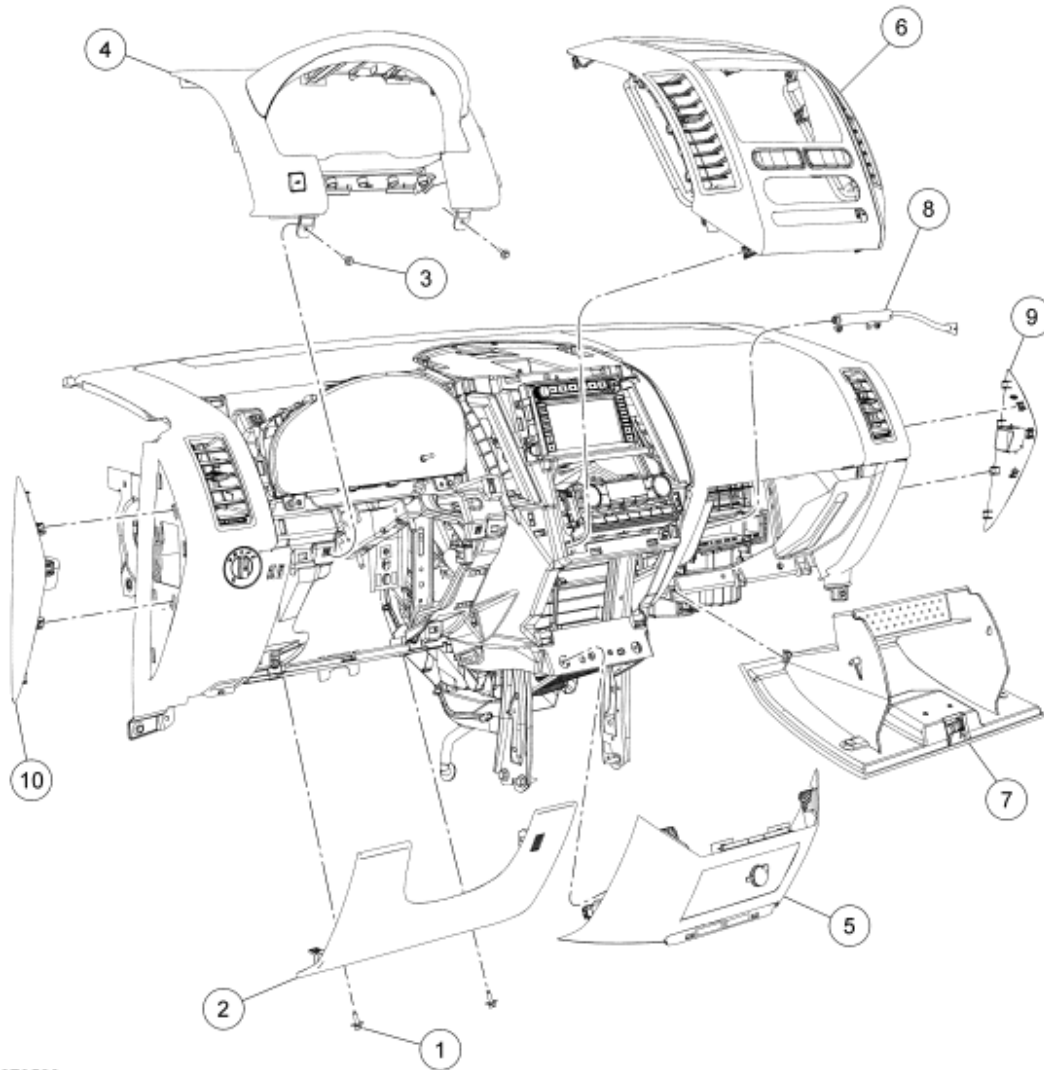
Description	Nm	lb-ft	lb-in
Floor console center bolts	6	-	53
Floor console front side bolts	6	-	53
Floor console rear bolts	6	-	53
Floor console-to-instrument panel bolts	6	-	53
Front of dash bolt	20	15	-
Instrument panel-to-steering column bracket bolts	22	16	-
LH instrument panel bolts	20	15	-
LH instrument panel-to-body bolt	7	-	62
LH instrument panel-to-floor bolts	22	16	-
RH instrument panel bolt	22	16	-
RH instrument panel-to-body bolt	7	-	62
RH instrument panel-to-floor bolt	22	16	-
Steering column pinch bolt	23	17	-
Torx head bolt	30	22	-

REMOVAL AND INSTALLATION

INSTRUMENT PANEL - EXPLODED VIEW

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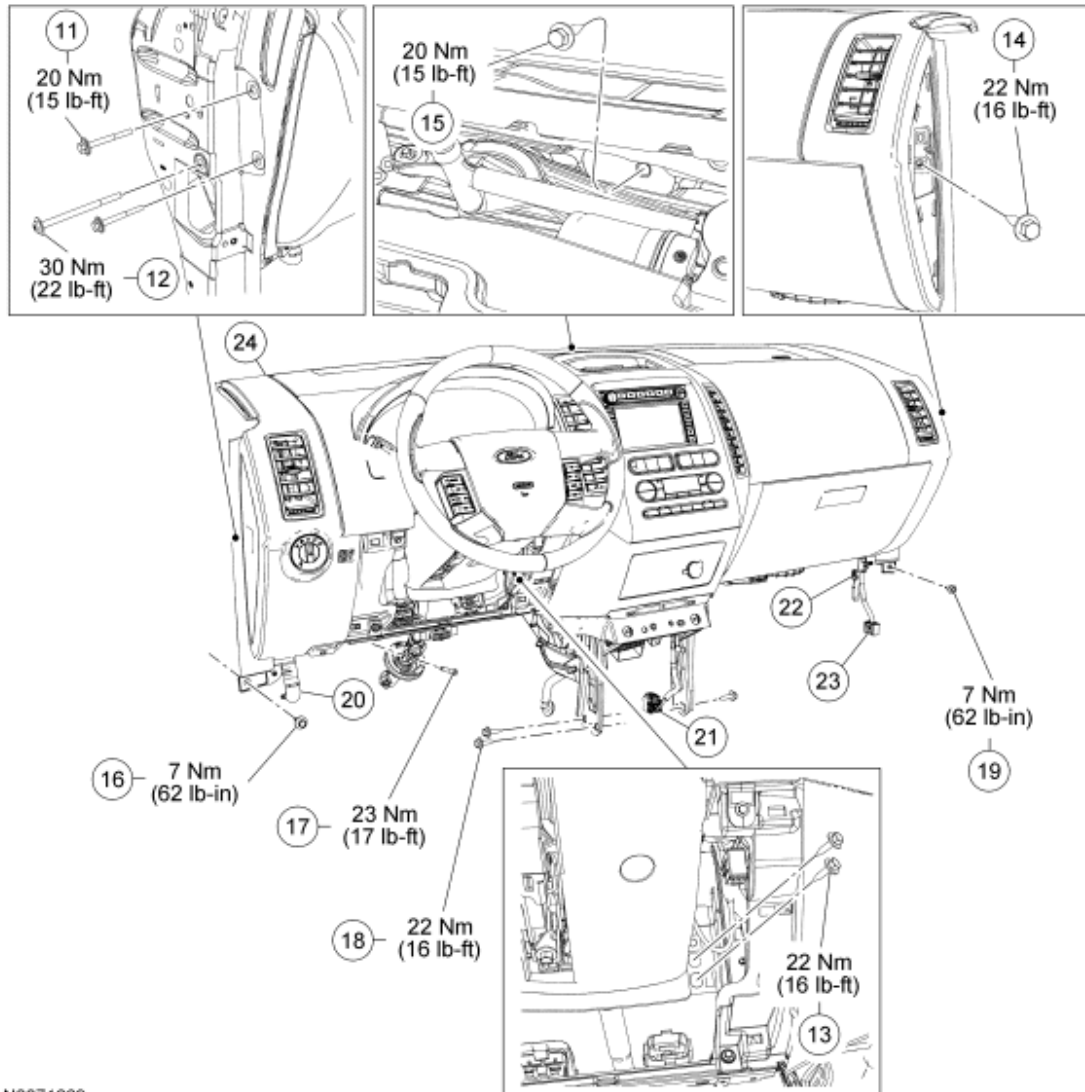
Fig. 1: Identifying Edge Instrument Panel (1 Of 2)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W505002	Steering column cover screw (2 required)
2	04459	Steering column opening cover
3	W707628	Instrument panel cluster finish panel screw (2 required)
4	044D70	Instrument panel cluster finish panel
5	04608	Instrument panel lower center finish panel
6	04302	Instrument panel upper center finish panel
7	06024	Glove compartment

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8	061A58	Glove compartment dampener
9	04480	RH instrument panel end trim panel
10	04481	LH instrument panel end trim panel



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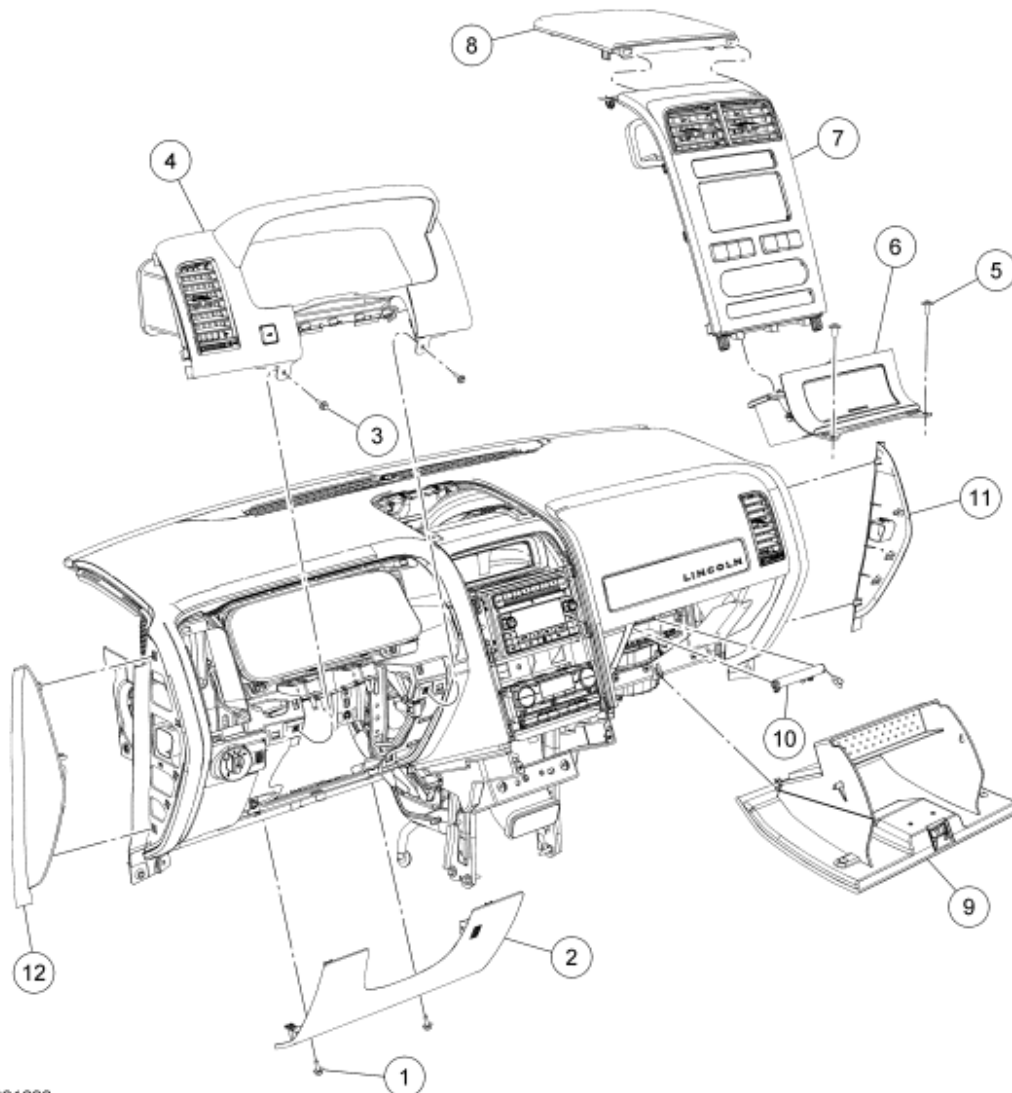
Fig. 2: Identifying Edge Instrument Panel (2 Of 2) With Torque Specifications
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
11	W712386	LH instrument panel bolt (2 required)
12	W712387	Torx head bolt
13	W711648	Instrument panel-to-steering column bracket bolt (2 required)
14	W706246	RH instrument panel bolt
15	W710309	Instrument panel-to-front of dash bolt

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16	W712403	LH instrument panel-to-body bolt
17	3RA27	Steering column pinch bolt
18	W706246	Instrument panel-to-floor bolt (3 required)
19	W706246	RH instrument panel-to-body bolt
20	14401	LH instrument panel electrical connectors
21	14401	Restraints control module (RCM) electrical connector
22	-	Antenna lead-in
23	14401	RH instrument panel electrical connector
24	04320	Instrument panel



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Fig. 3: Identifying MKX Instrument Panel (1 Of 2)

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Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W505002	Steering column cover screw (2 required)
2	04459	Steering column opening cover
3	W707628	Instrument panel cluster finish panel screw (2 required)
4	044D70	Instrument panel cluster finish panel
5	N811283	Instrument panel storage bin screw (2 required)
6	047A04	Instrument panel storage bin
7	04302	Instrument panel lower center finish panel
8	042A82	Instrument panel upper center finish panel
9	06024	Glove compartment
10	061A58	Glove compartment dampener
11	04480	RH instrument panel end trim panel
12	04481	LH instrument panel end trim panel

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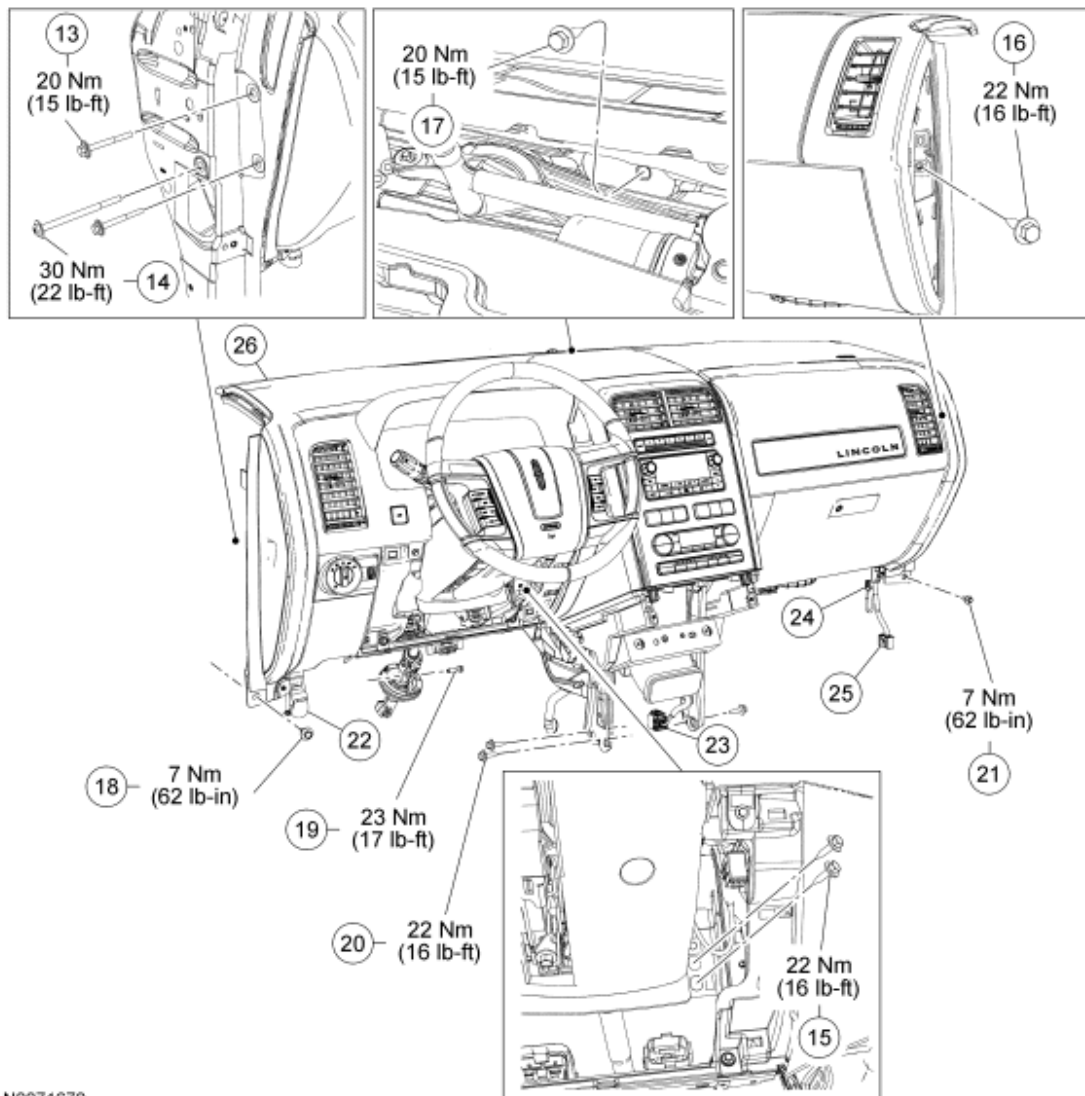


Fig. 4: Identifying MKX Instrument Panel (2 Of 2) With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
13	W712386	LH instrument panel bolt (2 required)
14	W712387	Torx head bolt
15	W711648	Instrument panel-to-steering column bracket bolt (2 required)
16	W706246	RH instrument panel bolt
17	W712403	Instrument panel-to-front of dash bolt
18	W703845	LH instrument panel-to-body bolt
19	3RA27	Steering column pinch bolt
20	W706246	Instrument panel-to-floor bolt (3 required)

21	W703845	RH instrument panel-to-body bolt
22	14401	LH instrument panel electrical connectors
23	14401	Restraints control module (RCM) electrical connector
24	-	Antenna lead-in
25	14401	RH instrument panel electrical connector
26	04320	Instrument panel

- For additional information, refer to the appropriate procedure.

INSTRUMENT PANEL CENTER FINISH PANEL

REMOVAL AND INSTALLATION

All vehicles

- Place the vehicle in NEUTRAL.

MKX only

- Remove the floor console trim ring.

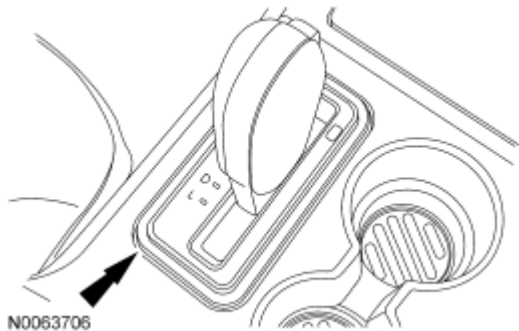


Fig. 5: Identifying Floor Console Trim Ring
Courtesy of FORD MOTOR CO.

All vehicles

- Remove the floor console finish panel.
 - Open the storage compartment door.
 - Remove the console finish panel.

MKX only

- Remove the storage bin.

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- Remove the 2 scrivenets.
- Disconnect the power point electrical connector.

NOTE: **The upper and lower center instrument panel finish panels are removed as one assembly.**

5. Using a suitable tool, gently pry out the upper instrument panel center finish panel to disengage the clips.
 - Disconnect the electrical connectors.
6. Using a suitable tool, gently pry out the lower instrument panel center finish panel to disengage the clips.
 - Disconnect the electrical connectors.
7. Separate the upper instrument panel finish panel from the lower instrument panel finish panel.

Edge only

NOTE: **The upper and lower instrument panel center finish panels are removed separately.**

8. Using a suitable tool, gently pry out the upper instrument panel center finish panel to disengage the clips and remove the panel.
 - Disconnect the electrical connectors.
9. Using a suitable tool, gently pry out the lower instrument panel center finish panel to disengage the clips and remove the panel.
 - Disconnect the electrical connectors.

All vehicles

10. To install, reverse the removal procedure.

INSTRUMENT CLUSTER FINISH PANEL

Edge only

1. Remove the upper instrument panel center finish panel.
 - Using a suitable tool, gently pry out the upper center finish panel to disengage the clips.
 - Remove the upper instrument panel center finish panel.
 - Disconnect the electrical connectors.

All vehicles

2. Lower the steering column to the full down position.
3. Remove the steering column opening cover.
 - Remove the 2 steering column opening cover screws.

- Disengage the steering column opening cover from the retainers.
- Disconnect the electrical connector and tube, if equipped.

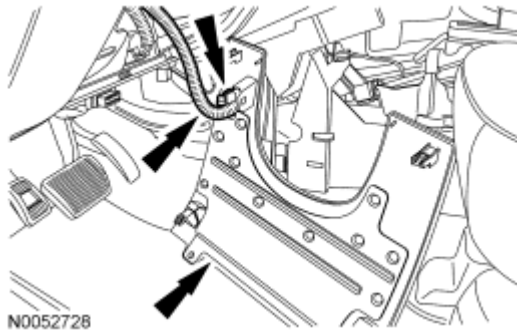


Fig. 6: Locating Electrical Connector, Tube & Steering Column Opening Cover
Courtesy of FORD MOTOR CO.

4. Remove the 2 cluster finish panel screws.

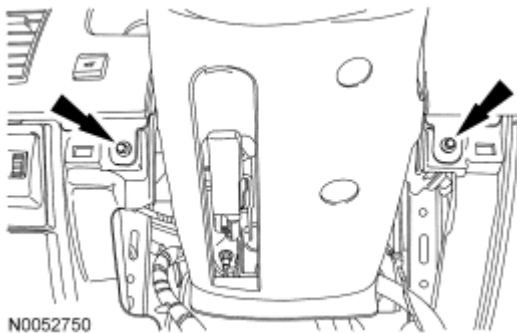


Fig. 7: Locating Cluster Finish Panel Screws
Courtesy of FORD MOTOR CO.

5. Disengage the instrument cluster finish panel from the retaining clips and remove the panel.
 - Disconnect the electrical connectors, if equipped.
6. To install, reverse the removal procedure.

INSTRUMENT PANEL

1. Remove the floor console. For additional information, refer to **Console - Floor**.
2. Depower the supplemental restraint system (SRS). For additional information, refer to **SUPPLEMENTAL RESTRAINT SYSTEM**.
3. Remove the weatherstripping from the front door openings near the instrument panel.
4. Remove the LH and RH instrument panel end trim panels.
5. Remove the LH and RH A-pillar trim panels. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION**.
6. Remove the LH and RH cowl panels.

7. Remove the steering column opening cover.
 - Remove the 2 steering column opening cover screws.
 - Disengage the steering column opening cover from the retaining clips.
 - Disconnect the electrical connector and hose, if equipped.

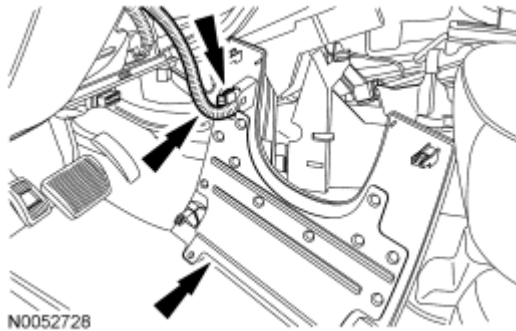


Fig. 8: Locating Electrical Connector, Tube & Steering Column Opening Cover
Courtesy of FORD MOTOR CO.

8. Remove the bolt and the hood opening handle.
 - Position the handle aside.

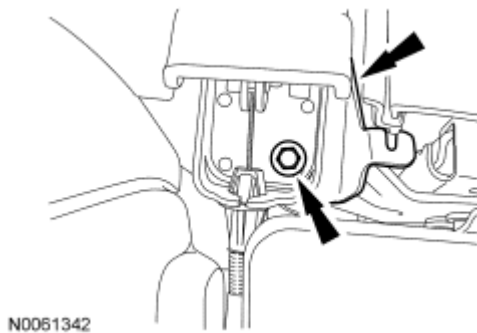


Fig. 9: Identifying Hood Opening Handle & Bolt
Courtesy of FORD MOTOR CO.

9. Disconnect the LH bulkhead electrical connector from under the steering column.

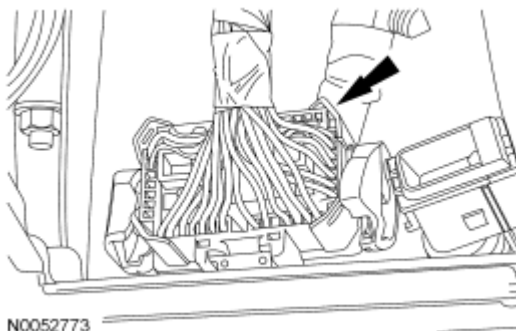


Fig. 10: Identifying LH Bulkhead Electrical Connector
Courtesy of FORD MOTOR CO.

10. Disconnect the RH bulkhead electrical connector from under the steering column.

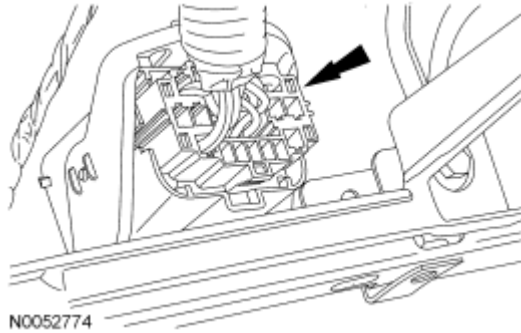


Fig. 11: Identifying RH Bulkhead Electrical Connector
Courtesy of FORD MOTOR CO.

11. Disconnect the 2 electrical connectors from the left cowl panel side.

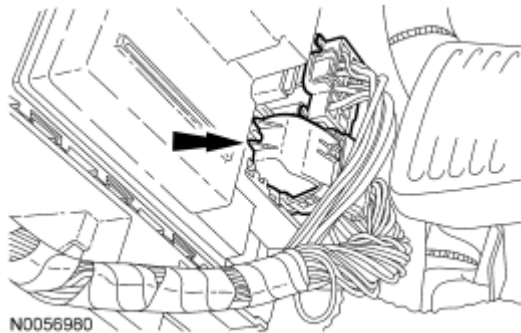


Fig. 12: Locating Electrical Connectors From Left Cowl Panel Side
Courtesy of FORD MOTOR CO.

12. Disconnect the electrical connectors and the antenna lead-in cable from the right cowl panel side.
- Remove the harness from the retainers.

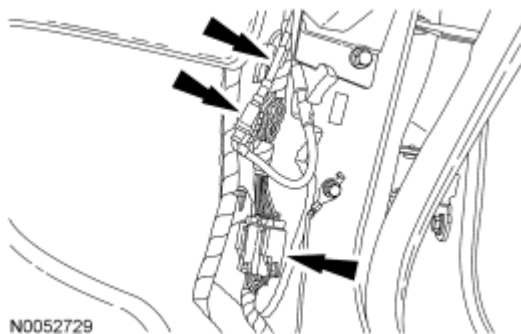


Fig. 13: Locating Electrical Connectors & Antenna Lead-In Cable From Right Cowl Panel Side

Courtesy of FORD MOTOR CO.

13. Disconnect the 2 A/C electrical connectors from inside the glove compartment.

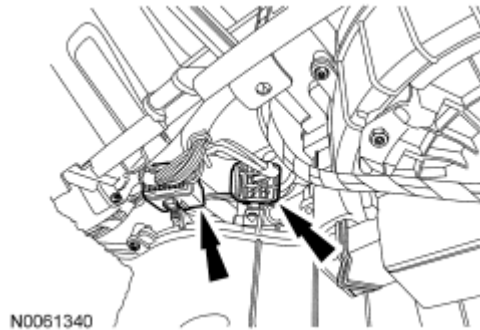


Fig. 14: Locating A/C Electrical Connectors From Inside Glove Compartment
Courtesy of FORD MOTOR CO.

14. Disconnect the electrical connector from the top of the heater core and evaporator core housing.

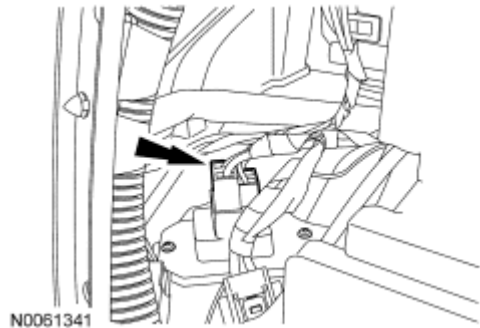


Fig. 15: Locating Blower Motor Resistor Or Blower Motor Speed Control Electrical Connector
Courtesy of FORD MOTOR CO.

15. Disconnect the selector lever cable from the ball stud on the selector lever.

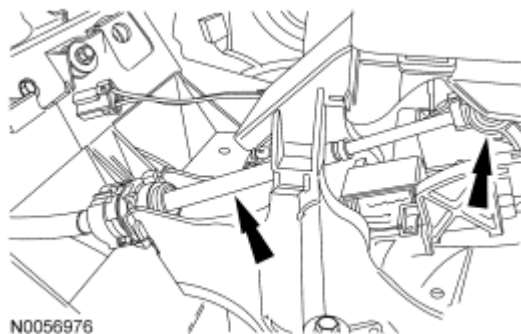


Fig. 16: Locating Shift Cable From Ball Stud On Shifter
Courtesy of FORD MOTOR CO.

16. Remove the selector lever cable from the retaining clips.

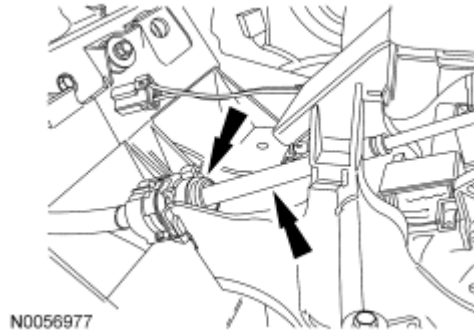


Fig. 17: Locating Shift Cable From Retaining Clips
Courtesy of FORD MOTOR CO.

17. Remove the 4 screws (2 each side) from the inner floor console support.

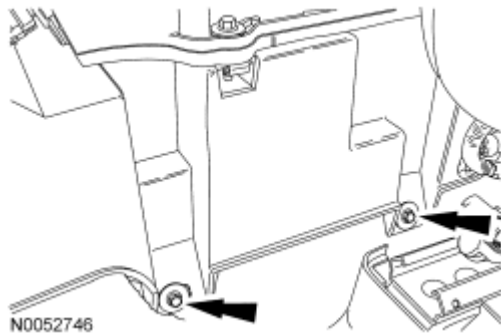


Fig. 18: Locating Screws From Inner Floor Console Support
Courtesy of FORD MOTOR CO.

18. Remove the screws from the front inner floor console support-to-instrument panel.

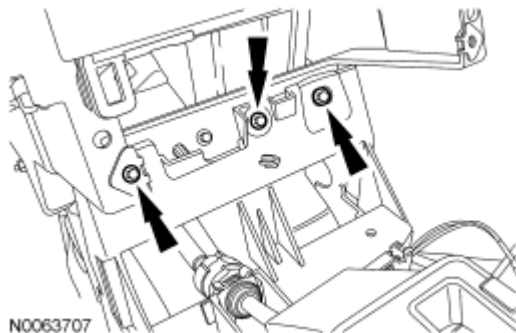


Fig. 19: Locating Screws From Front Inner Floor Console Support-To-Instrument Panel
Courtesy of FORD MOTOR CO.

19. Disconnect the electrical connectors from the front inner floor console.

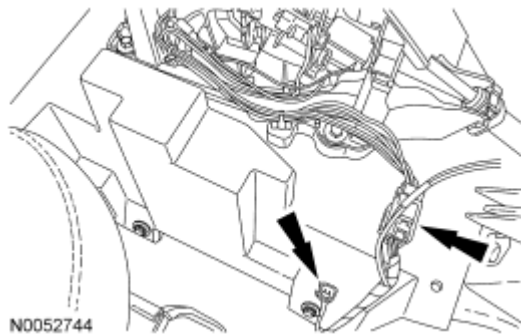


Fig. 20: Locating Electrical Connectors From Front Inner Floor Console
Courtesy of FORD MOTOR CO.

20. Remove the inner floor console support.
21. Disconnect the small restraints control module (RCM) electrical connector.

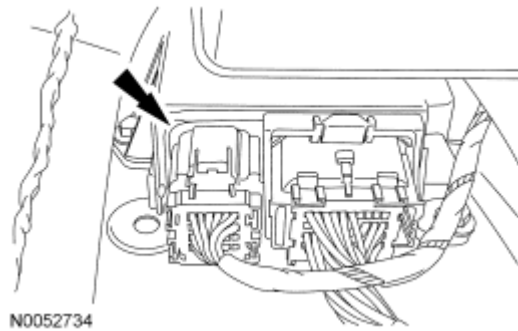


Fig. 21: Locating Restraints Control Module (RCM) Small Electrical Connector
Courtesy of FORD MOTOR CO.

22. Remove the LH pushpin retainer and position the carpet aside.

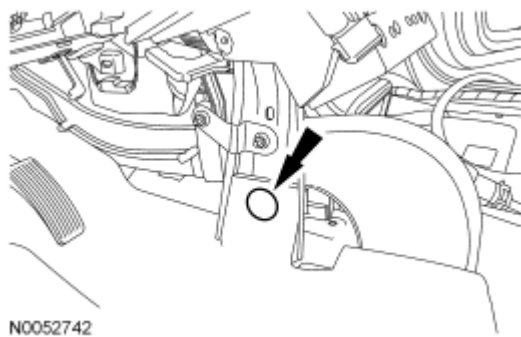


Fig. 22: Locating LH Push Pin Retainer & Position Carpet Aside
Courtesy of FORD MOTOR CO.

23. Remove the LH pushpin retainer from the instrument panel-to-A/C duct.

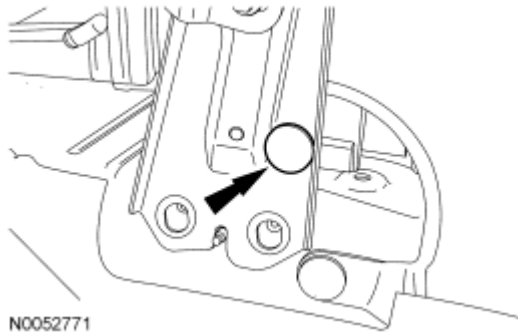


Fig. 23: Locating LH Push Pin Retainer From Instrument Panel-To-A/C Duct
Courtesy of FORD MOTOR CO.

24. Remove the 2 LH instrument panel-to-floor bolts.
 - To install, tighten to 22 N.m (16 lb-ft).
25. Remove the RH pushpin retainer and position the carpet aside.

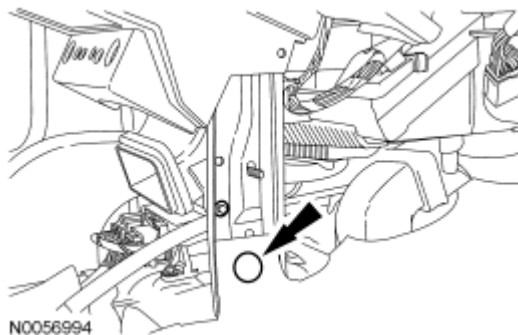


Fig. 24: Locating LH Instrument Panel-To-Floor Bolts
Courtesy of FORD MOTOR CO.

26. Remove the screw from the RH instrument panel brace-to-A/C duct.

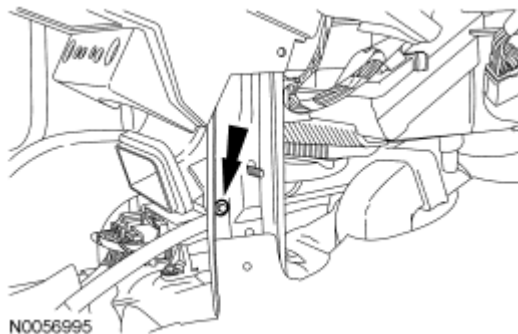


Fig. 25: Locating Screw From RH Instrument Panel Brace-To-A/C Duct
Courtesy of FORD MOTOR CO.

27. Remove the RH instrument panel-to-floor bolt.

- To install, tighten to 22 N.m (16 lb-ft).

28. Remove the 2 nuts and the LH instrument panel brace-to-A/C duct.

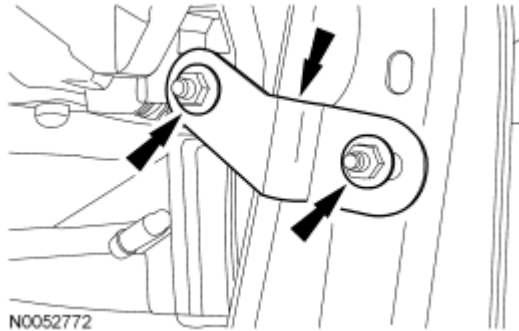


Fig. 26: Locating LH Instrument Panel Brace-To-A/C Duct Nuts
Courtesy of FORD MOTOR CO.

29. Remove and discard the steering column pinch bolt.

- To install, use a new steering column pinch bolt and tighten to 23 Nm (17 lb-ft).

30. Rotate the steering wheel until the front wheels are in the straight-ahead position.

- Remove the ignition key.

CAUTION: Do not allow the steering column shaft to rotate while the intermediate shaft is disconnected or damage to the clockspring may result. If there is evidence that the shaft has rotated, the clockspring must be removed and recentered. For additional information, refer to SUPPLEMENTAL RESTRAINT SYSTEM .

31. Detach the steering column shaft from the intermediate shaft.

32. Remove the upper cowl panel grille. For additional information, refer to FRONT END BODY PANELS .

33. Remove the bolt from the front of dash.

- To install, tighten to 20 N.m (15 lb-ft).

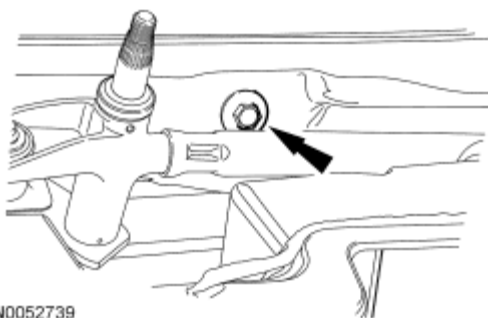


Fig. 27: Locating Bolt From Front Of Dash
Courtesy of FORD MOTOR CO.

34. Remove the driver seat. For additional information, refer to SEATING .

NOTE: The Torx head bolt is not completely removed. The bolt is pulled outward far enough to allow for clearance when removing the instrument panel.

35. Remove the Torx head bolt from inside the LH door.
- To install, tighten to 30 N.m (22 lb-ft).

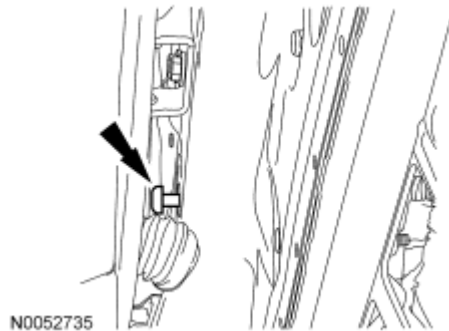


Fig. 28: Locating Torx(R) Head Bolt From Inside LH Door
Courtesy of FORD MOTOR CO.

36. Remove the 2 LH instrument panel bolts from inside the LH door.
- To install, tighten to 20 N.m (15 lb-ft).

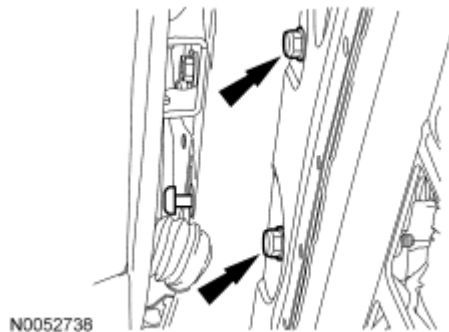


Fig. 29: Locating Bolts From Inside LH Door
Courtesy of FORD MOTOR CO.

37. Remove the 2 instrument panel-to-steering column bracket bolts from the bracket under the steering column.
- To install, tighten to 22 N.m (16 lb-ft).

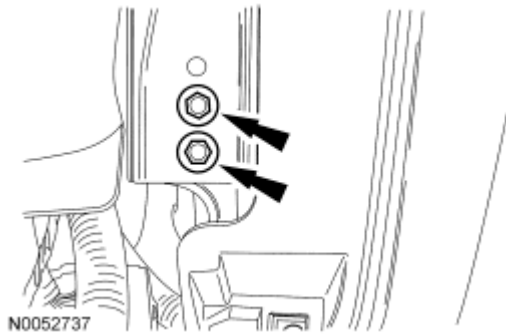


Fig. 30: Identifying Bolts From Bracket Under Steering Column
Courtesy of FORD MOTOR CO.

38. Remove the LH instrument panel-to-body bolt.
- To install, tighten to 7 N.m (62 lb-in).

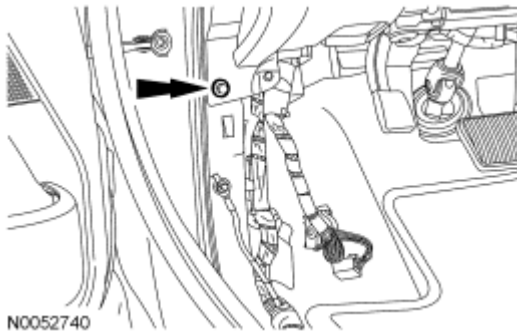


Fig. 31: Identifying LH Instrument Panel-To-Body Bolt
Courtesy of FORD MOTOR CO.

39. Remove the bolt from the RH instrument panel.
- To install, tighten to 22 N.m (16 lb-ft).

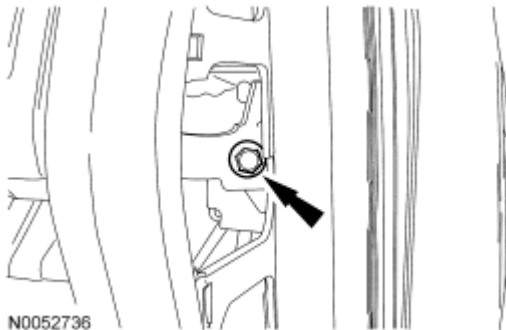
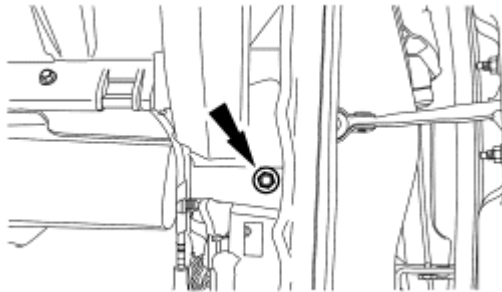


Fig. 32: Locating Bolt From RH Instrument Panel Cowl Side Panel
Courtesy of FORD MOTOR CO.

40. Remove the RH instrument panel-to-body bolt.

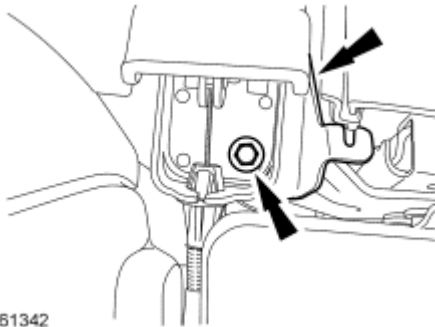
- To install, tighten to 7 N.m (62 lb-in).



N0061339

Fig. 33: Identifying RH Instrument Panel-To-Body Bolt
Courtesy of FORD MOTOR CO.

41. Remove the bolt and the hood opening handle.
 - Position the handle aside.



N0061342

Fig. 34: Identifying Hood Opening Handle & Bolt
Courtesy of FORD MOTOR CO.

CAUTION: To avoid damage to the instrument panel, this next step requires the help of an assistant.

42. Remove the instrument panel.
43. To install, reverse the removal procedure.

POWER POINT

Special Tools

Illustration	Tool Name	Tool Number
	Remover, Power Point Socket	501-039 or equivalent

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Instrument Panel & Console - Edge & MKX

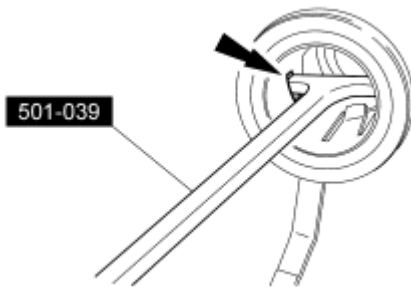


ST3024-A

REMOVAL

NOTE: Power point cover may differ depending on location.

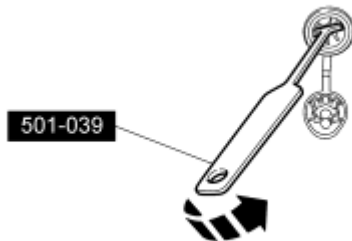
1. Open the power point cover.
2. Install the Power Point Socket Remover in one of the power point socket slots.



N0061767

Fig. 35: Installing Special Tool In Power Point Socket Slot
Courtesy of FORD MOTOR CO.

3. Position the Power Point Socket Remover so that it engages in the adjacent slot.



N0061768

Fig. 36: Positioning Special Tool To Engage In Adjacent Slot
Courtesy of FORD MOTOR CO.

4. Using the Power Point Socket Remover, pull the power point socket out of the retainer.

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Instrument Panel & Console - Edge & MKX

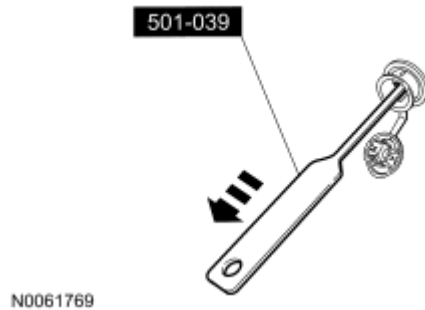


Fig. 37: Pulling Power Point Socket Out Of Retainer
Courtesy of FORD MOTOR CO.

5. Disconnect the electrical connector.

INSTALLATION

1. Connect the electrical connector.
2. Slide the power point socket into the retainer.

CONSOLE - FLOOR

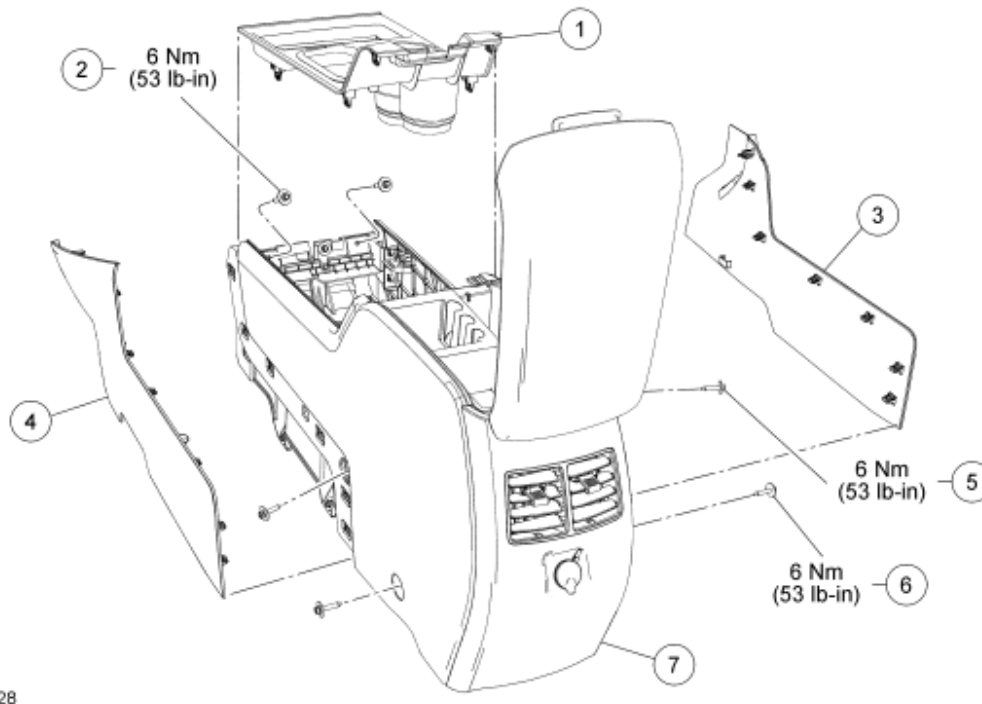


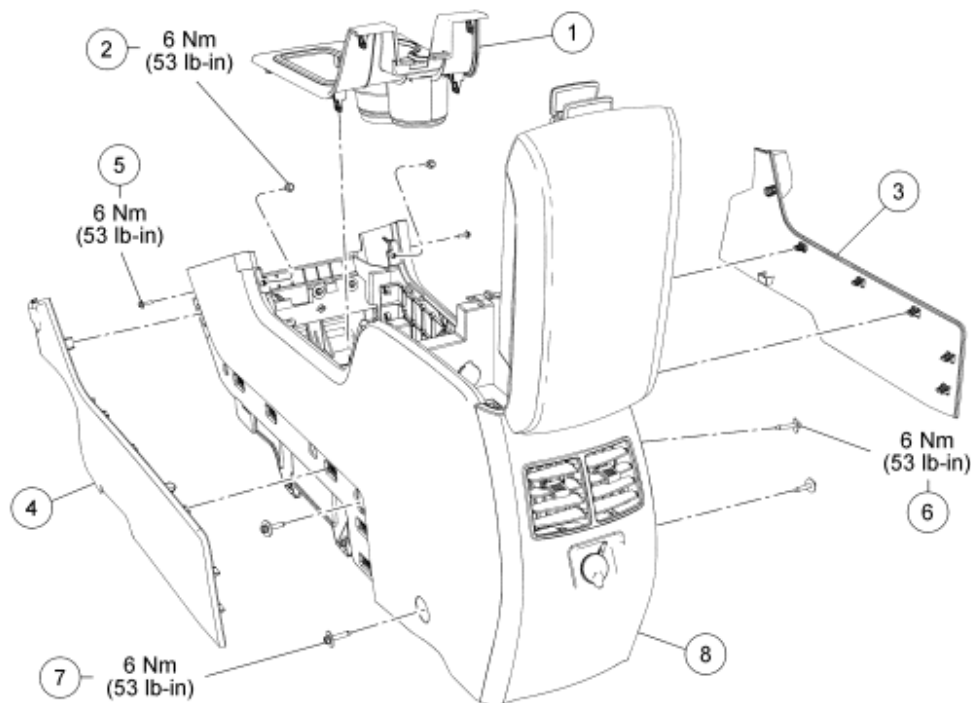
Fig. 38: Identifying Edge Floor Console With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
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2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Instrument Panel & Console - Edge & MKX

1	045A76	Floor console finish panel
2	W706862	Floor console-to-instrument panel bolt (2 required)
3	04608	Floor console RH side panel
4	04609	Floor console LH side panel
5	W706862	Floor console center bolt (2 required)
6	W706862	Floor console rear bolt (2 required)
7	045A36	Floor console



N0061429

Fig. 39: Identifying MKX Floor Console With Torque Specifications
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	045A76	Floor console finish panel
2	W706862	Floor console-to-instrument panel bolt (2 required)
3	04608	Floor console RH side panel
4	04609	Floor console LH side panel
5	W707628	Floor console front side bolt (2 required)
6	W706862	Floor console center bolt (2 required)
7	W706862	Floor console rear bolt (2 required)
8	045A36	Floor console

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Instrument Panel & Console - Edge & MKX

All vehicles

1. Position the selector lever to NEUTRAL.

NOTE: **The seats must be in the full raised position to access the bolts.**

2. Position the seats forward and remove the 2 floor console rear bolts.
 - To install, tighten to 6 N.m (53 lb-in).
3. Remove the 2 floor console side panels.

NOTE: **The seats must be in the full raised position to access the bolts.**

4. Position the seats rearward and remove the 2 floor console side panels and 2 floor console center bolts.
 - To install, tighten to 6 N.m (53 lb-in).
5. Remove the floor console finish panel.

MKX only

6. Remove the storage bin.
 - Remove the 2 scrivers.
 - Disconnect the power point electrical connector.
7. Remove the 2 floor console front side bolts.
 - To install, tighten to 6 N.m (53 lb-in).

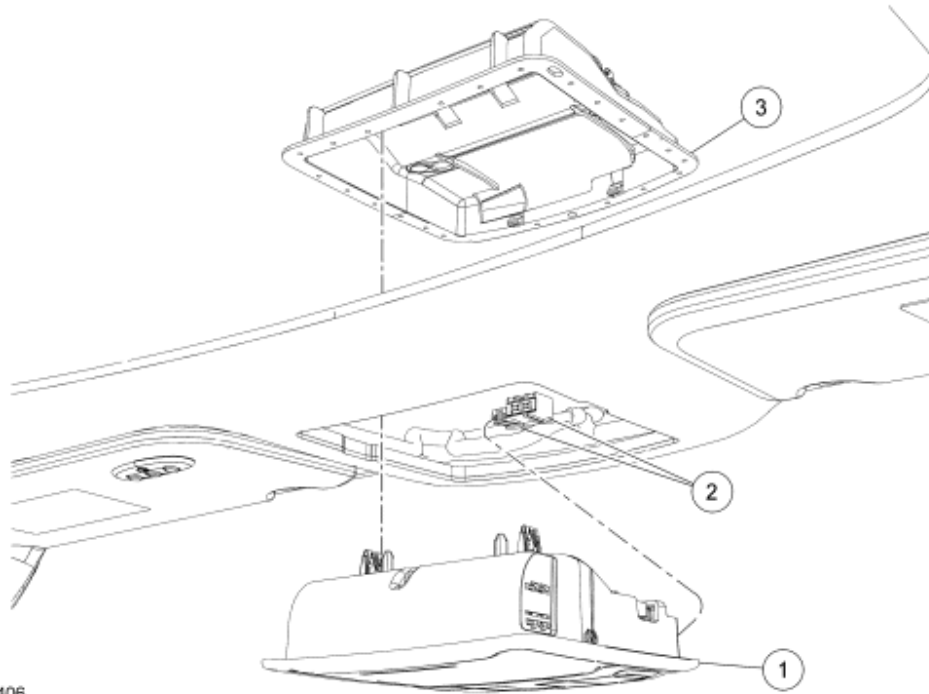
All vehicles

8. Remove the 2 floor console-to-instrument panel bolts.
 - To install, tighten to 6 N.m (53 lb-in).
9. Disconnect the floor console electrical connector from the lower right front wiring harness.
10. Remove the floor console.
11. To install, reverse the removal procedure.

CONSOLE - OVERHEAD

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Instrument Panel & Console - Edge & MKX



N0061406

Fig. 40: Exploded View Of Overhead Console
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	519A70	Overhead console
2	-	Overhead console electrical connectors
3	519K22	Overhead console bracket

1. Remove the overhead console.
 - Pull downward on the overhead console.
 - Disconnect the electrical connectors.
2. To install, reverse the removal procedure.